

RETAILPULSE BI: EXECUTIVE BUSINESS PERFORMANCE REPORT

Enterprise Sales Performance, Profitability Audit & Strategic Decision Briefing

Prepared for: Chief Executive Officer (CEO), Senior Leadership | Date: September 2026 | Domain: BI Analytics

1. Executive Summary

Project Scope: RetailPulse Commerce completed a comprehensive business intelligence audit analyzing 4,995 transaction records across 12 calendar months (2025). The goal of this project was to transition the enterprise from manual spreadsheet reliance to an automated, high-performance Data Analytics and BI Command Center.

TOTAL REVENUE	TOTAL NET PROFIT	NET PROFIT MARGIN
\$12,584,567.30	\$3,765,203.73	29.92%
TOTAL ORDERS	UNITS SOLD	AVG ORDER VALUE
4,995 Orders	11,172 Units	\$2,519.43

2. Business Scenario & Objectives

Operational Context: RetailPulse Commerce operates a multi-channel retail infrastructure selling across five product categories (Electronics, Fashion, Home & Kitchen, Sports, Beauty), utilizing two sales channels (Online and Physical Store) across four geographic territories (West, South, North, East).

- Centralized Reporting:** Eliminate fragmented spreadsheets and establish a single source of truth for financial KPIs.
- Margin Leakage Identification:** Pinpoint specific products generating high sales volume but experiencing margin erosion due to over-discounting.
- Regional & Channel Optimization:** Quantify revenue contribution vs. transaction volume across geographic sales territories and e-commerce channels.
- Executive Briefing:** Provide actionable insights, strategic recommendations, and answer the CEO Question Challenge.

3. Data Quality Audit & Transformation

Quality Assurance: A rigorous data cleaning workflow was executed across the raw transactional dataset (5 sheets, 4,995 rows) prior to relational database loading and Power BI modeling.

- **Deduplication & Order Audit:** Verified 4,995 unique transaction IDs with zero duplicate order rows.
- **Geography Dimension Cleaning:** Identified and removed 8 trailing blank-region rows in the raw Geography sheet, deduplicating on City to yield 36 unique primary cities.
- **Date Table Expansion:** Extracted Month (MMM) and Year-Month (YYYY-MM) attributes from Order_Date to enable temporal trend analysis.
- **Formula Standardization:** Recalculated Revenue = Quantity * Unit_Price * (1 - Discount), Profit = Revenue - Cost, and Margin % = Profit / Revenue.

4. SQL Analytical Engine & Category Matrix

Database Aggregation: The cleaned dataset was imported into MySQL (database: retailpulse) to perform multi-dimensional aggregations, customer DENSE_RANK() segmentation, and category profitability evaluations.

Product Category	Orders	Units Sold	Total Revenue (\$)	Total Profit (\$)	Profit Margin (%)
Electronics	1,190	2,697	\$3,696,120.55	\$1,278,198.32	34.58%
Fashion	1,008	2,226	\$3,139,889.00	\$825,379.72	26.29%
Home & Kitchen	1,206	2,635	\$2,487,036.93	\$730,933.33	29.39%
Sports	631	1,421	\$1,639,327.58	\$492,672.39	30.05%
Beauty	960	2,193	\$1,622,193.26	\$438,019.98	27.00%

5. Star Schema Data Architecture & DAX

Data Modeling: A Star Schema data model was constructed in Power BI Desktop linking 1 Fact table (Orders) with 4 Dimension tables (Customers, Products, Calendar, Geography). All relationships enforce 1-to-many (1:*) cardinality with single-directional cross-filtering.

- **Total Revenue DAX:** SUMX(Orders, Orders[Quantity] * Orders[Unit_Price] * (1 - Orders[Discount]))
- **Total Profit DAX:** SUMX(Orders, (Orders[Quantity] * Orders[Unit_Price] * (1 - Orders[Discount])) - Orders[Cost])
- **Profit Margin DAX:** DIVIDE([Total Profit], [Total Revenue], 0) -- Evaluates to 29.92%
- **Average Order Value DAX:** DIVIDE([Total Revenue], [Total Orders], 0) -- Evaluates to \$2,519.43

6. Key Business Insights & Analytical Callouts

INSIGHT #1 — MARGIN LEAKAGE IN TECH ACCESSORIES

Key Finding: USB-C Hub (\$688K Rev) and Backpack (\$541K Rev) rank in the top 10 revenue drivers, but suffer from low profit margins of 20.65% and 18.78% (baseline: 29.92%).

Business Implication: Aggressive promotional discounting (>20%) and vendor cost inflation are eating into net margins on high-volume SKUs.

Strategic Action: Cap maximum promotional discounts on accessories at 10% and renegotiate wholesale supplier unit pricing.

INSIGHT #2 — REGIONAL REVENUE DEFICIT IN EAST TERRITORY

Key Finding: The West region generates \$4.34M (34.5% share) of total revenue, whereas the East region generates only \$1.90M (15.1% share).

Business Implication: East underperformance is strictly driven by transaction volume deficit (758 orders vs West's 1,716 orders) rather than unit margin (East margin is 30.24%).

Strategic Action: Reallocate digital marketing ad spend to target major East region metro markets (Kolkata, Patna, Bhubaneswar).

INSIGHT #3 — HIGH-MARGIN SUPERSTAR PRODUCTS

Key Finding: Mechanical Keyboard (\$996K Rev, 41.00% Margin) and Smartwatch (\$540K Rev, 44.05% Margin) generate premium profits across all product lines.

Business Implication: Premium hardware exhibits strong consumer price inelasticity and low discount sensitivity.

Strategic Action: Feature high-margin flagship products prominently on the online homepage and offer hardware bundle options.

7. The CEO Question Challenge (3 Key Takeaways)

Executive Briefing: If presenting this Business Intelligence dashboard directly to the Chief Executive Officer, the following three core data-driven points summarize the state of the business:

THE CEO BRIEFING: 3 THINGS THE CEO MUST KNOW

1. Revenue vs. Margin Asymmetry: Total Revenue reached \$12.58M with \$3.77M Profit (29.92% Margin). However, top-volume accessories like USB-C Hub (\$688K Rev) and Backpack (\$541K Rev) suffer from weak margins (~18%-20%) due to heavy discounting, whereas premium hardware (Mechanical Keyboard @ 41% margin, Smartwatch @ 44% margin) drives overall gross profit.

2. Geographic Sales Imbalance: The West Region dominates with \$4.34M (34.5% share) of total revenue, whereas the East Region generates only \$1.90M (15.1% share). East underperformance is driven strictly by lower transaction volume (758 orders vs West's 1,716 orders) despite maintaining a healthy 30.24% profit margin.

3. E-Commerce Channel Dominance: The Online Sales Channel commands 61.7% (\$7.76M) of total company revenue with a higher Average Order Value (\$2,529.26 vs In-Store's \$2,503.76), proving that consumers prefer digital purchasing for high-ticket tech and fashion items.

8. Strategic Recommendations & Roadmap

- **Month 1 — Discount Governance:** Implement hard discount caps (max 10%) on high-demand tech accessories in the ERP system.
- **Month 2 — East Region Ad Push:** Launch targeted regional ad campaigns in East metro cities to boost order frequency.
- **Month 3 — E-Commerce Bundles:** Deploy e-commerce bundle promotions (Keyboard + Mouse) to increase average basket size.
- **Ongoing — BI Automation:** Connect Power BI Service to automated daily MySQL refresh pipelines for real-time monitoring.

Financial Impact: By executing these recommendations, RetailPulse Commerce can capture an estimated \$350,000+ in annual profit recovery while expanding customer transaction volume across underperforming regional markets.